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Liaw

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(54) **SEMICONDUCTOR DEVICE LAYOUT**

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H01L 21/762	(2006.01)

(Continued)

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(58) **Field of Classification Search**

CPC . H10D 64/017; H10D 84/853; H10D 84/0172

See application file for complete search history.

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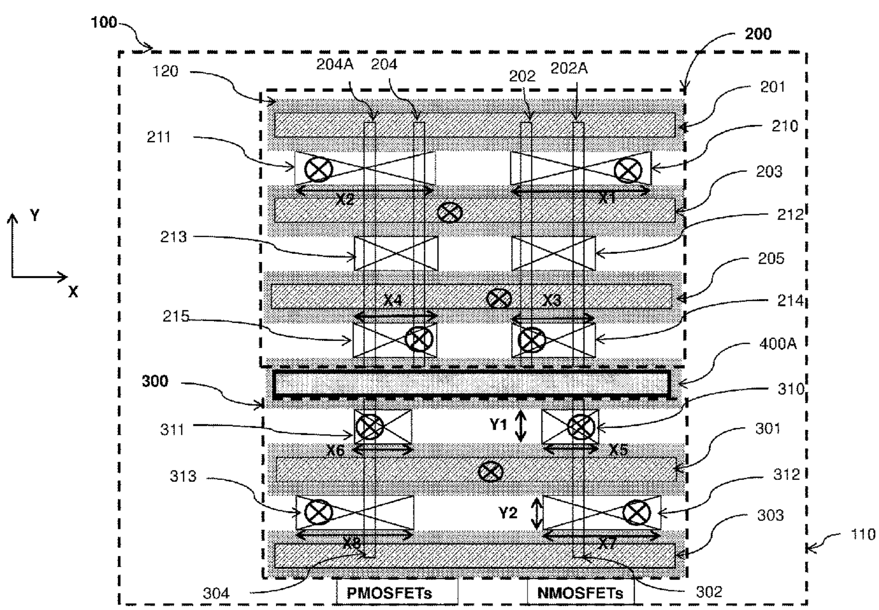
(57) **ABSTRACT**

Semiconductor devices and semiconductor cell arrays are provided herein. In some examples, a semiconductor device includes a multi-fin active region, a mono-fin active region, and an isolation feature between the multi-fin active region and the mono-fin active region. The multi-fin active region includes a first plurality of fins, a second plurality of fins parallel to the first plurality of fins, a first n-type field effect transistor (FET), and a first p-type FET. The mono-fin active region abuts the multi-fin active region. The mono-fin active region includes a first fin, a second fin different from the first fin, a second n-type FET, and a second p-type FET. The isolation feature is parallel to the first and second gate structures.

20 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/721,197, filed on Dec. 19, 2019, now Pat. No. 11,063,032, which is a division of application No. 15/718,696, filed on Sep. 28, 2017, now Pat. No. 10,522,528.

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H10D 84/85	(2025.01)
H10D 89/10	(2025.01)
H01L 21/3105	(2006.01)
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H10D 62/834	(2025.01)
H10D 64/62	(2025.01)
H10D 64/66	(2025.01)
H10D 64/68	(2025.01)

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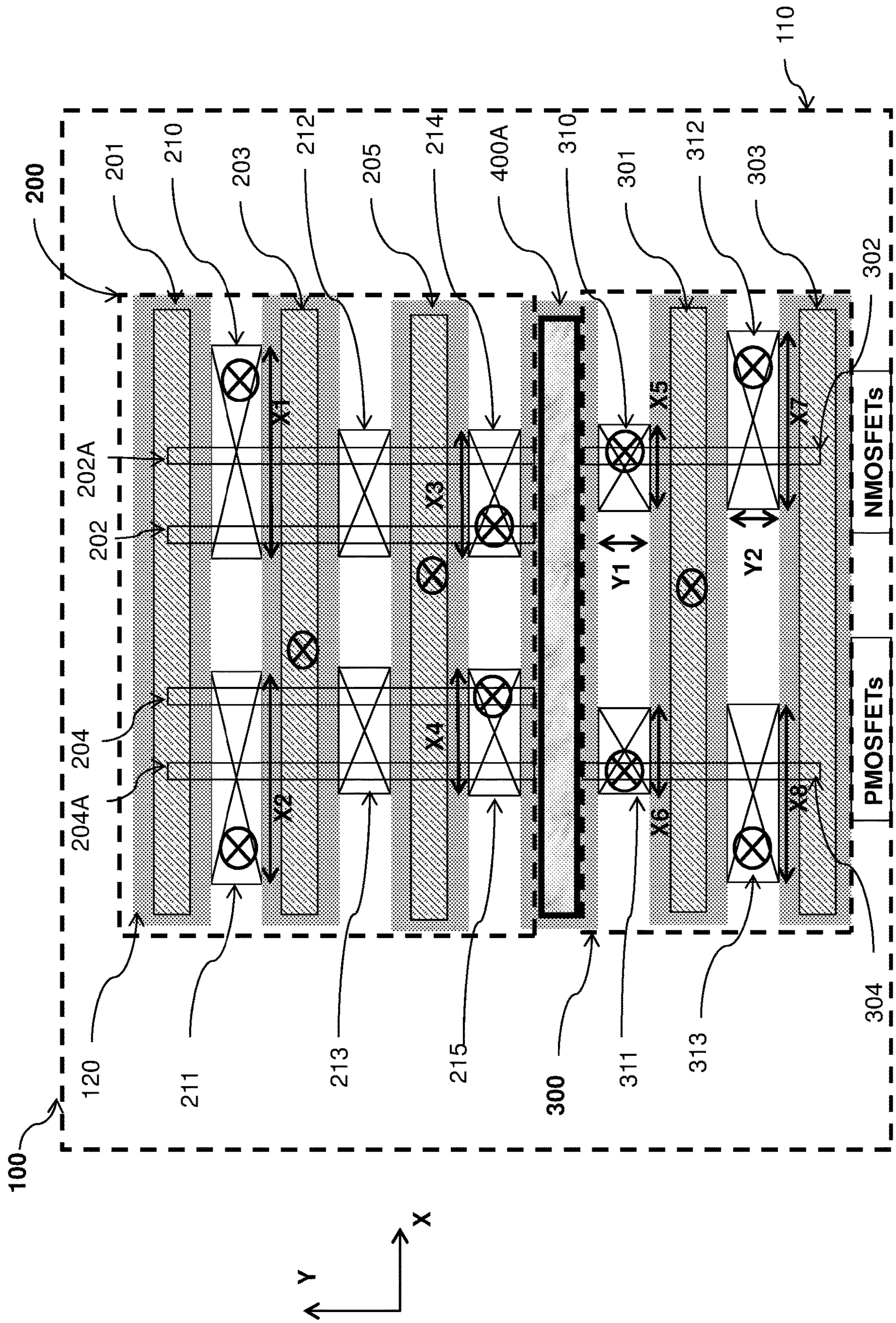
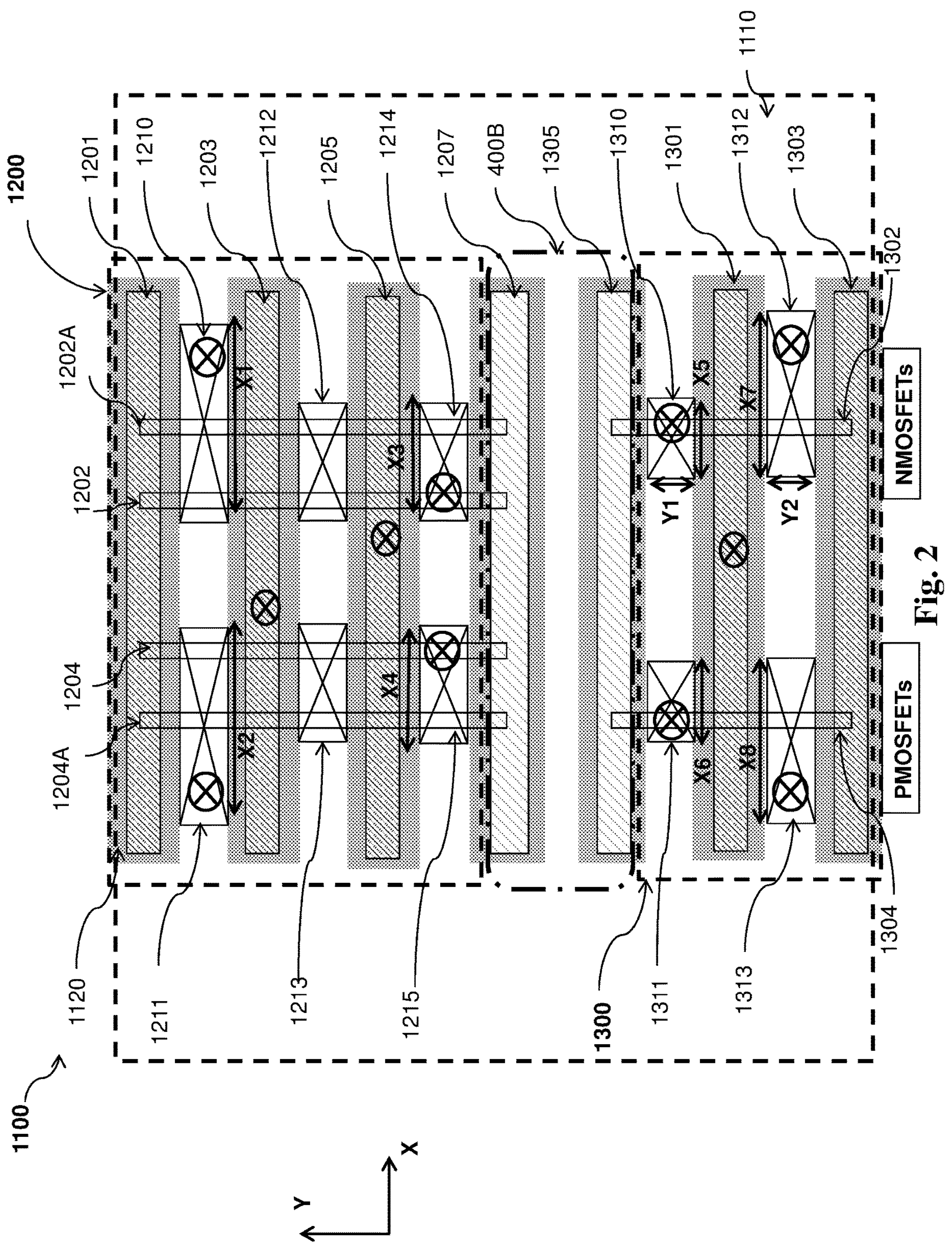


100
110
120
200
201
202
202A
203
204
204A
205
210
211
212
213
214
215
300
301
302
303
304
310
311
312
313
400A
X1
X2
X3
X4
X5
X6
X7
X8
Y1
Y2
NMOSFETs
PMOSFETs
X
Y

Fig. 1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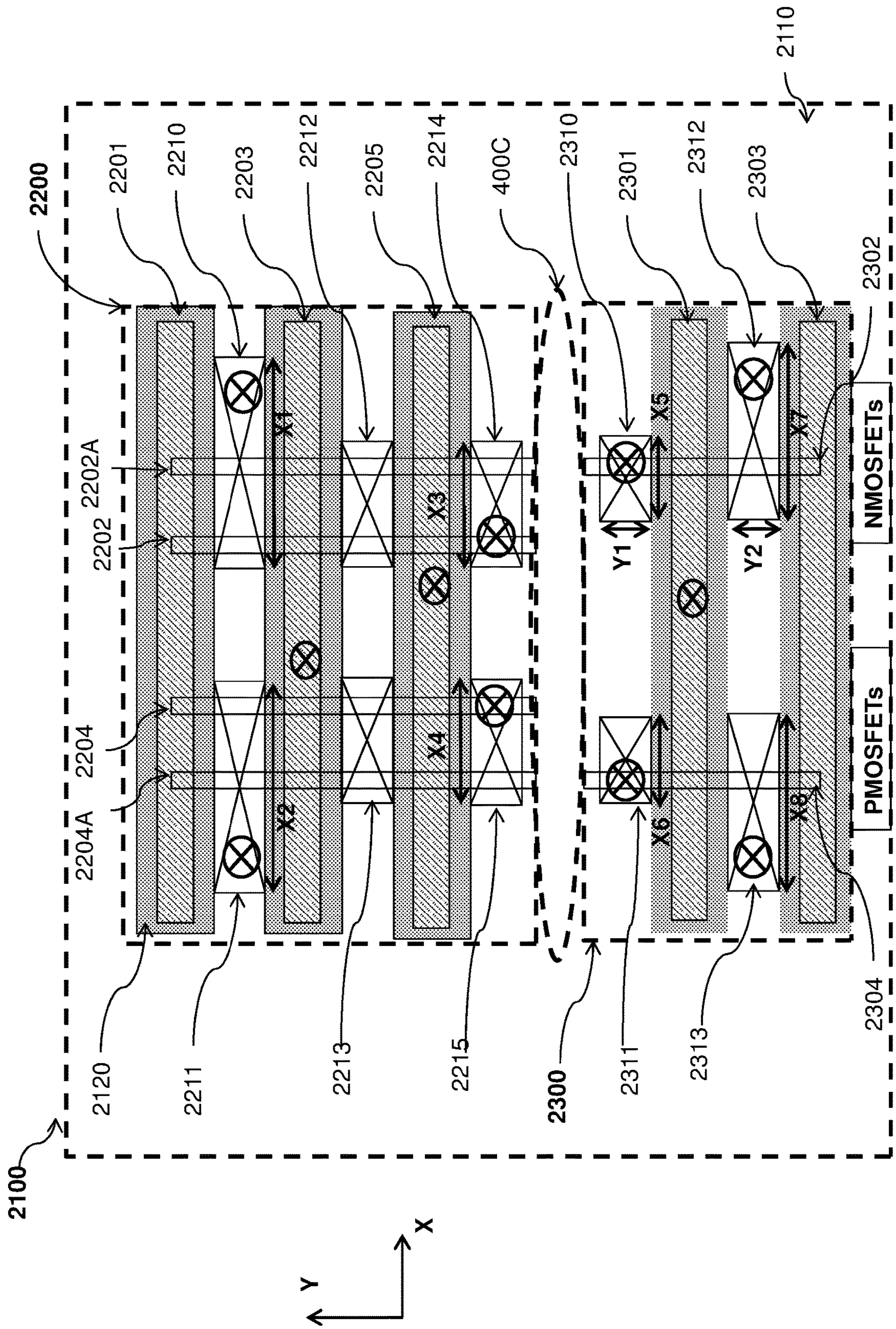
1214
1215
400B
1300
1301
1302
1303
1304
1305
1310
1311
1312
1313
X1
X2
X3
X4
X5
X6
X7
X8
Y1
Y2
NMOSFETs
PMOSFETs
X
Y

Fig. 2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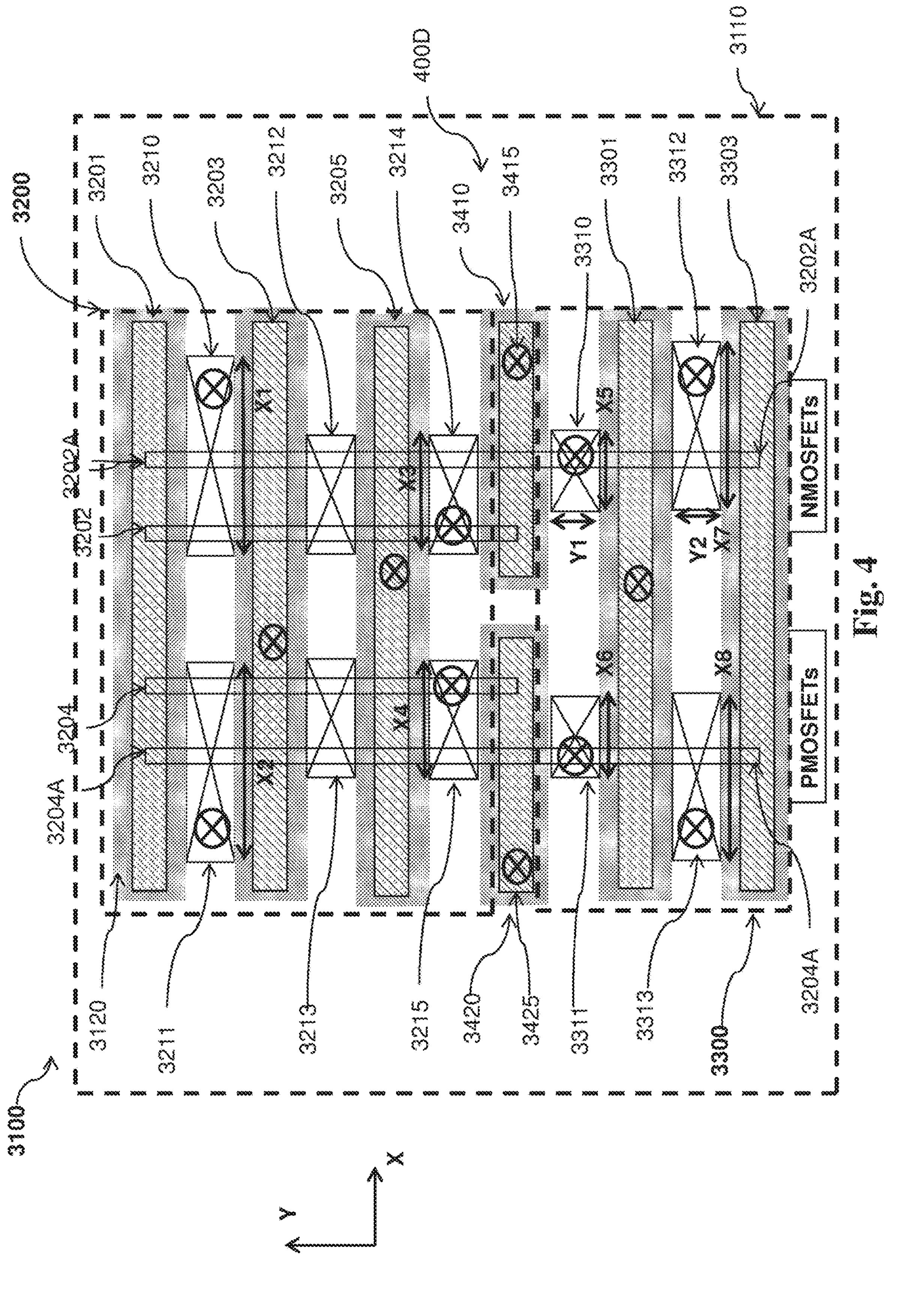
2214
2215
400C
2301
2302
2303
2304
2310
2311
2312
2313
X1
X2
X3
X4
X5
X6
X7
X8
Y1
Y2
NMOSFETs
PMOSFETs
X
Y

Fig. 3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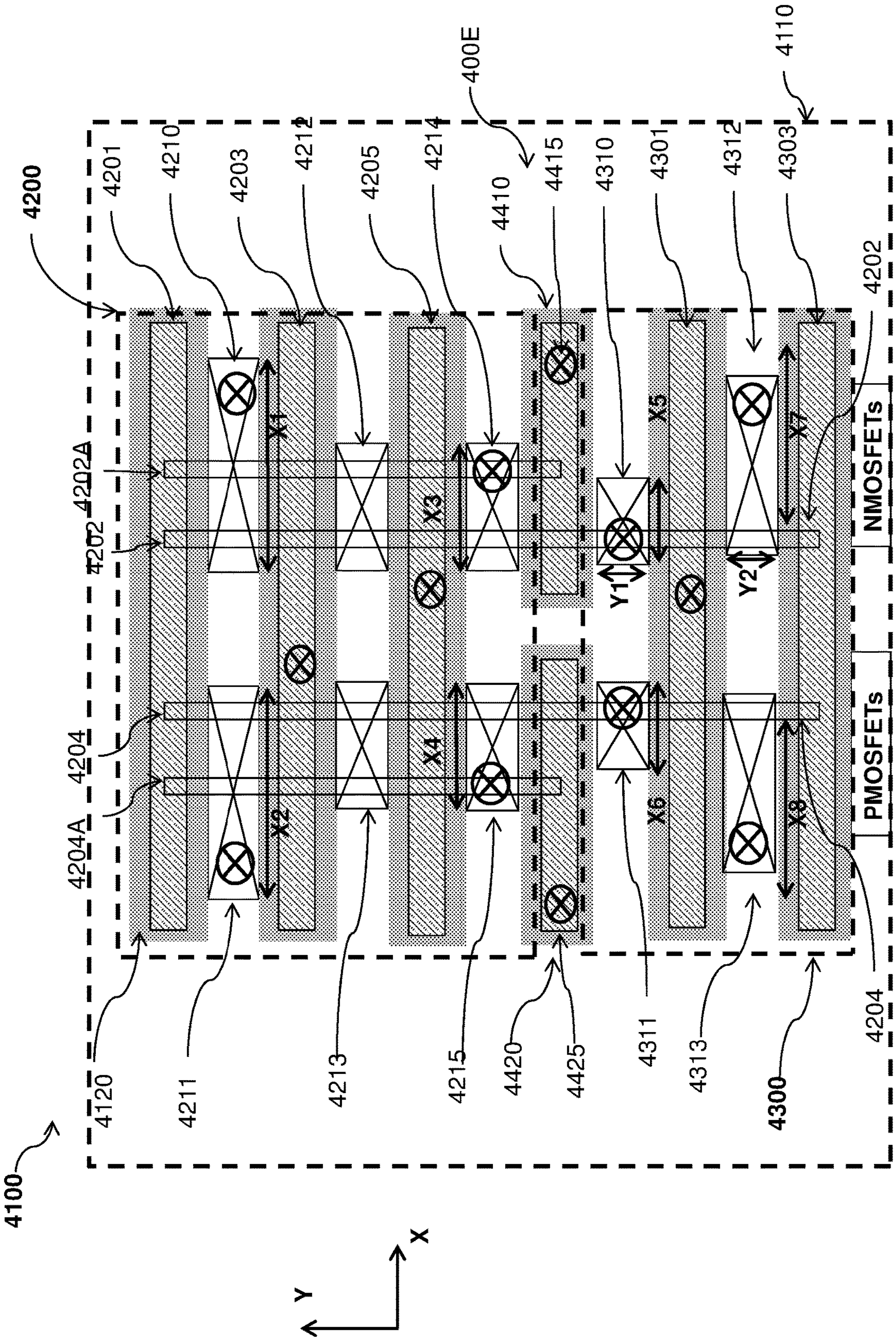
3202
3204
3204A
3120
3211
3213
3215
3420
3425
3311
3313
3300
X1
X2
X3
X4
X5
X6
X7
X8
Y1
Y2
NMOSFETs
PMOSFETs
X
Y

Fig. 4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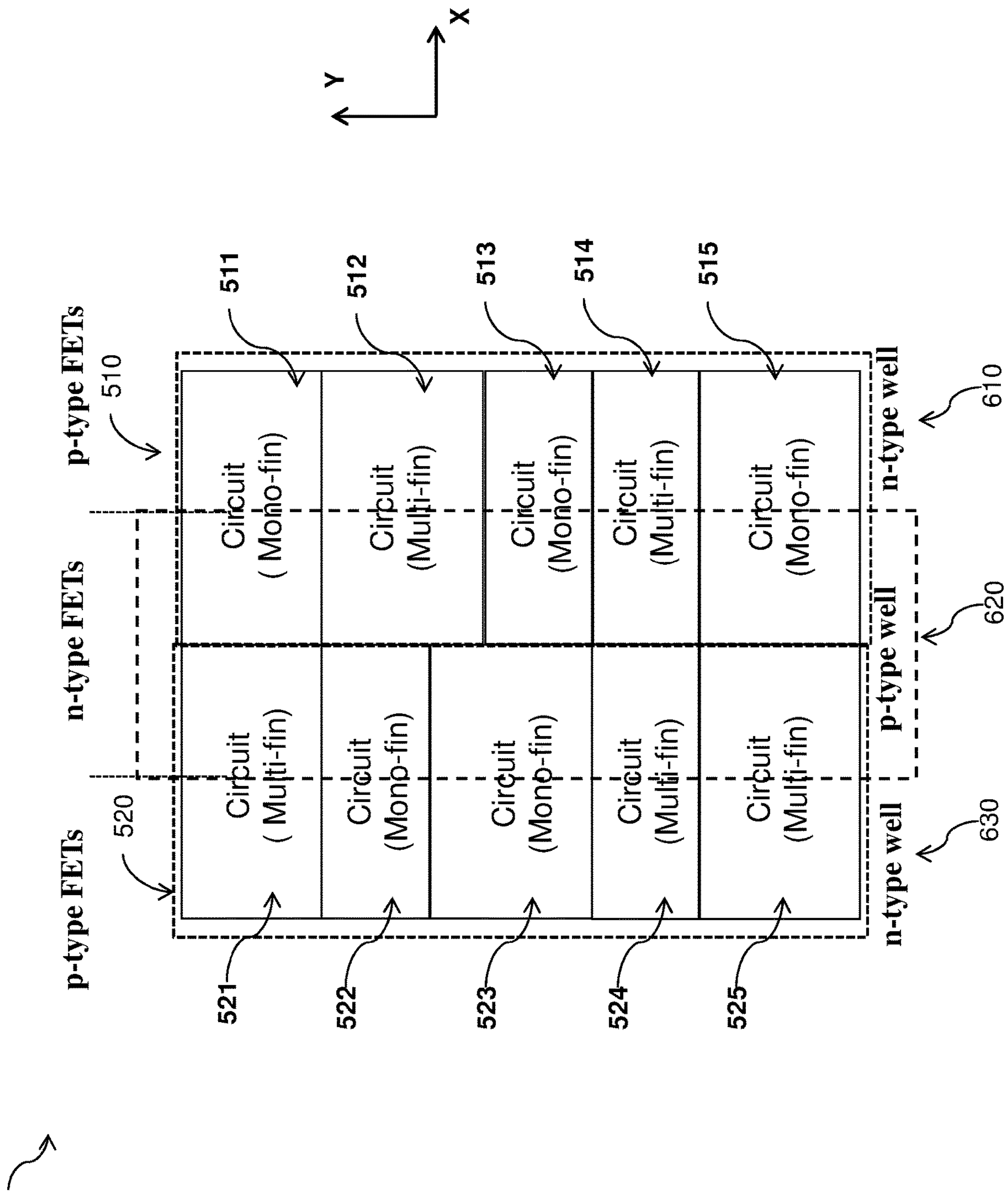
4303
4202
4202A
4204
4204A
4211
4213
4215
4420
4425
4311
4313
4300
X1
X2
X3
X4
X5
X6
X7
X8
Y1
Y2
NMOSFETs
PMOSFETs
X
Y

Fig. 5


X
Y
p-type FETs
510
n-type FETs
p-type FETs
520
511
512
513
514
515
Circuit (Mono-fin)
Circuit (Multi-fin)
Circuit (Mono-fin)
Circuit (Multi-fin)
Circuit (Mono-fin)
Circuit (Multi-fin)
Circuit (Mono-fin)
Circuit (Mono-fin)
Circuit (Multi-fin)
Circuit (Multi-fin)
521
522
523
524
525
n-type well
610
p-type well
620
n-type well
630

500

Fig. 6

SEMICONDUCTOR DEVICE LAYOUT

PRIORITY DATA

This is a continuation application of U.S. patent application Ser. No. 17/373,302, filed Jul. 12, 2021, which is a continuation application of U.S. patent application Ser. No. 16/721,197, filed Dec. 19, 2019, now issued as U.S. Pat. No. 11,063,032, which is divisional application of U.S. patent application Ser. No. 15/718,696, filed Sep. 28, 2017, now issued as U.S. Pat. No. 10,522,528, each of which is incorporated herein by reference in its entirety.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs, but it has also increased the complexity of processing and manufacturing ICs.

For example, fin-like field effect transistors (FinFETs) have become a popular choice for design of high performance circuitry. While FinFETs' narrow fin width helps achieve short channel control, their source/drain (S/D) features tend to have a small landing for low-contact-resistance S/D contacts. FinFETs with multiple fins, or multi-fin FETs, are proposed for high-speed applications. However, multi-fin FETs suffer higher leakage and therefore higher power consumption when compared with FinFETs with a single fin.

Accordingly, improvements in semiconductor devices to achieve both high switching speed and low power consumption are desired.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** is a diagrammatic top view of a semiconductor device, according to aspects of the present disclosure.

FIG. **2** is a diagrammatic top view of a semiconductor device layout, according to aspects of the present disclosure.

FIG. **3** is a diagrammatic top view of a semiconductor device, according to aspects of the present disclosure.

FIG. **4** is a diagrammatic top view of a semiconductor device, according to aspects of the present disclosure.

FIG. **5** is a diagrammatic top view of a semiconductor device, according to aspects of the present disclosure.

FIG. **6** is a diagrammatic top view of a semiconductor cell array, according to aspects of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The present disclosure is generally related to semiconductor devices, and more particularly, related to standard cells that include multi-fin devices or mono-fin devices.

Referring to FIG. **1**, illustrated therein is a top view of a semiconductor device **100** according to aspects of the present disclosure. The semiconductor device **100** includes silicon-containing fins **202**, **202**A, **204**, **204**A, **302**, and **304**. Fins **202**, **202**A, **204**, and **204**A are formed in a multi-fin active region **200** while fins **302** and **304** are formed within a mono-fin active region **300**. In some instances, fins **202** and **202**A are formed over a p-type well on semiconductor substrate **110** and fins **204** and **204**A are formed over an n-type well on semiconductor substrate **110**. In some embodiments, upon completion of the multi-fin active region **200**, fins **202** and **202**A are part of at least one n-type FET (nFET) and fins **204** and **204**A are part of at least one p-type FET (pFET). While FIG. **1** shows that each of the nFET and pFET in multi-fin active region **200** includes two fins, implementation with more than two fins per FET can be appreciated by people skilled in the art upon examination of the present disclosure. In cases where each FETs of multi-fin active region **200** include two fins, the multi-fin active region can be referred to as a double-fin active region. In some embodiments, each of multi-fin active region **200** and mono-fin active region **300** constitutes a standard cell. In that regard, multi-fin active region **200** can be referred to as a multi-fin standard cell and mono-fin active region **300** can be referred to as a mono-fin standard cell.

In some embodiments, fin **302** is formed over the same p-type well where fins **202** and **202**A are formed and fin **304** is formed over the same n-type well where fins **204** and **204**A are formed. Upon completion of the mono-fin active region **300**, fin **302** is part of an nFET and fin **304** is part of a pFET. As their names suggest, each the nFET and pFET in multi-fin active region **200** includes more than one fin while each of the nFET and pFET in mono-fin active region **300** includes a single fin. The semiconductor substrate **110** usually includes silicon. Alternatively, the semiconductor substrate **110** may include thereover epitaxial layers of germanium, silicon germanium, or other semiconductor materials and combinations. In some instances, depending on the design the semiconductor device **100**, the semiconductor substrate **110** may be doped with p-type dopants such as boron (B), aluminum (Al) and gallium (Ga) or n-type dopants such as antimony (Sb), arsenic (As) and phosphorous (P).

In some embodiments, a fin such as fins **202**, **202**A, **204**, **204**A, **302** and **304** is formed of epitaxial layers on the semiconductor substrate **110** and the epitaxial layers are formed of silicon (Si) alone or together with a semiconductor material that is compatible with silicon. Such a semiconductor material includes germanium (Ge) and carbon (C). In some implementations, the epitaxial layers can include layers of different compositions. In some instances, the epitaxial layers include alternating layers of two different compositions. Introduction of Ge or C into Si lattice is known to strain the Si lattice and is usually utilized to improve the device performance in certain aspects. In some embodiments, the epitaxial layers are formed of epitaxial growth of Si, C and Ge and combinations thereof using techniques such as epitaxial deposition by chemical vapor deposition (CVD) or low-pressure chemical vapor deposition (LPCVD). By controlling the delivery of gas reactants and other process parameters during the CVD epitaxial deposition, the concentrations of Si, C and/or Ge along the height of the epitaxial layers can be modulated. In embodiments where the fins **202**, **202**A, **204**, **204**A, **302** and **304** are formed of epitaxial layers, the epitaxial layers are first formed over the semiconductor substrate **110** and then the epitaxial layers are patterned as described below. In some embodiments, regions for n-type FETs and P-type FETs are independently tuned for enhanced electron-mobility and hole-mobility, respectively. For examples, silicon carbide and silicon germanium are epitaxially grown in the regions for N-type FETs and the regions for P-type FETs, respectively. In some other examples, carbide and germanium are doped into the regions for N-type FETs and the regions for P-type FETs, respectively by ion-implantation. Further, in some implementations, the epitaxial layers on the semiconductor substrate **110** can also be doped to p-type dopants such as B, Al, Ga or n-type dopants such as Sb, As, and P. In those implementations, the resulting fins **202**, **202**A, **204**, **204**A, **302** and **304**, as the case may be, would be doped accordingly.

In some embodiments, the fins **202**, **202**A, **204**, **204**A, **302**, and **304** are formed from the epitaxial layers by photolithography patterning and etching. For example, a patterned photoresist layer is formed on the epitaxial layers by a photolithography technique, and then an etching process, such as anisotropic etching, is applied to the epitaxial layers to form one or more fins. In another example, a hard mask is used. In that case, the hard mask is formed by depositing a hard mask material on the epitaxial layer. A photoresist layer is then deposited on the hard mask. After patterned using photolithography, the photoresist on the hard mask then serves as the etch mask when the hard mask is etched and patterned. Thereafter, an etching process, such as anisotropic etching, is applied to the epitaxial layers to form one or more fins using the hard mask as an etch mask. To isolate a fin from an adjacent fin, a dielectric material (such as thermally grown silicon oxide and CVD deposited silicon oxide) is formed to fill trenches between a fin and its neighboring fins. The dielectric layer is then polished by chemical mechanical polishing (CMP) and then etched back to expose a portion of the fin while a portion of the fin remains covered by the etched back dielectric layer, usually referred to as shallow trench isolation (STI). For example, fins **202** and **202**A, fins **202** and **204**, and fins **204** and **204**A are each isolated from one another by STI features. In some embodiments, fin **204**A and **304** are first formed as a unitary fin before further processes separate them. In some other embodiment, fin **204** is first formed as part of a fin that extends into mono-fin active region **300** before later processes remove the extension of fin **204** in mono-fin active region **300**. The same applies to fins **202** and **202**A. Fins **202**A and **302** are first formed as a continuous fin before they are separated at a later process step. Fin **202** is formed across both multi-fin active region **200** and mono-fin active region **300** before its extension into mono-fin active region **200** is removed at a later operation.

As shown in FIG. **1**, in some embodiments, multi-fin active region **200** includes gate structure **201**, **203** and **205** and mono-fin active region **300** includes gate structures **301** and **303**. Gate structures **201**, **203** and **205** are formed over and span across fins **202**A, **202**, **204**, and **204**A. Gate structures **301** and **303** are formed over and span across fins **302** and **304**. To form these gate structures, dummy gates are first formed at their current locations and then these dummy gates are replaced by high-K metal gate stack. The formation of a dummy gate includes depositing a dummy gate layer containing polysilicon (poly-Si) or other suitable material and patterning the dummy gate layer. A gate hard mask layer may be formed on the dummy gate material layer and is used as an etch mask during the formation of the dummy gate. The gate hard mask layer may include any suitable material, such as a silicon oxide (SiO_2), a silicon nitride (SiN), a silicon carbide (SiC), a silicon oxynitride (SiON), other suitable materials, and/or combinations thereof. In some embodiments, the patterning process to form a dummy gate includes forming a patterned resist layer by lithography process; etching the hard mask layer using the patterned resist layer as an etch mask; and etching the gate material layer to form the dummy gate using the patterned hard mask layer as an etch mask.

To form functional gate structure **201**, **203**, **205**, **301**, and **303**, dummy gates are replaced with high-K metal gate stacks. In some embodiments, the high-K metal gate stack at least includes a gate dielectric layer interfacing the fins and a metal layer (not shown) over the gate dielectric layer. The gate dielectric layer can be formed of high-K dielectrics such as hafnium oxide (HfO_2), zirconium oxide (ZrO_2), tantalum oxide (Ta_2O_5), barium titanate ($BaTiO_3$), titanium dioxide (TiO_2), cerium oxide (CeO_2), lanthanum oxide (La_2O_3), lanthanum aluminum oxide ($LaAlO_3$), lead titanate ($PbTiO_3$), strontium titanate ($SrTiO_3$), lead zirconate ($PbZrO_3$), tungsten oxide (WO_3), yttrium oxide (Y_2O_3), bismuth silicon oxide ($Bi_4Si_2Oi_2$), barium strontium titanate (BST) ($Ba_{1-x}Sr_xTiO_3$), PMN ($PbMg_xNb_{1-x}O_3$), PZT ($PbZr_xTi_{1-x}O_3$), PZN ($PbZn_xNb_{1-x}O_3$), and PST ($PbSc_xTa_{1-x}O_3$), lead lanthanum titanate, strontium bismuth tantalate, bismuth titanate and barium zirconium titanate. In some instances, the high-K metal gate stack may include one or more work function metal layers formed of, for example, TiN, TaN, TaCN, TiCN, TiC, Mo, and W.

In some instances, one or more gate sidewall features (or gate spacers) **120** are formed on the sidewalls of gate structure **201**, and similarly on the sidewalls of the other gate structures **203**, **205**, **301** and **303**. The gate spacers **120** may be used to offset the subsequently formed S/D features and may be used for designing or modifying the S/D feature profile. The gate spacers **120** may include any suitable dielectric material, such as a semiconductor oxide, a semiconductor nitride, a semiconductor carbide, a semiconductor oxynitride, other suitable dielectric materials, and/or combinations thereof. The gate spacers **120** may have multiple layers, such as two layers (a silicon oxide (SiO_2) film and a silicon nitride (SiN) film) or three layers (a silicon oxide

(SiO_2) film; a silicon nitride (SiN) film; and a silicon oxide (SiO_2) film). The formation of the gate spacers **120** includes deposition and anisotropic etching, such as dry etching.

Source/drain (S/D) features are formed over the fins on either side of a non-floating gate structure, such as gate structure **203**, **205** and **301**. As shown in FIG. **1**, for the nFET controlled by gate structure **203** in multi-fin active region **200**, a S/D feature is formed below S/D contact **210** and a S/D feature is formed below S/D contact **212**, with S/D contacts **210** and **212** on different sides of gate structure **203**. In some implementations, with respect to any FET in multi-fin active region **200**, a S/D feature and its corresponding S/D contact are formed across and span over all fins of that FET. For example, S/D contact **210** and the S/D feature therebelow and S/D contact **212** and the S/D feature therebelow are formed over and span across fins **202** and **202**A. Similarly, for the nFET controlled by gate structure **205** in multi-fin active region **200**, S/D contact **212** and the S/D feature therebelow are on one side of gate structure **205** while S/D contact **214** and the S/D feature therebelow are on the other side of gate structure **205**. The same applies to the pFETs in multi-fin active region **200** and the nFET and pFET in mono-fin active region **300**. For the pFET controlled by gate structure **203**, S/D contact **211** and the S/D feature therebelow are on one side of gate structure **203** while S/D contact **213** and S/D feature therebelow are on the other side of gate structure **203**. For the pFET controlled by gate structure **205**, S/D contact **213** and the S/D feature therebelow are on one side of gate structure **205** while S/D contact **215** and S/D feature therebelow are on the other side of gate structure **205**. For the nFET controlled by gate structure **301**, S/D contact **310** and the S/D feature therebelow are on one side of gate structure **301** while S/D contact **312** and S/D feature therebelow are on the other side of gate structure **301**. Finally, for the pFET controlled by gate structure **301**, S/D contact **311** and the S/D feature therebelow are on one side of gate structure **301** while S/D contact **313** and S/D feature therebelow are on the other side of gate structure **301**.

At least for purpose of this disclosure, an FET that includes multiple fins is still considered one FET as long as its S/D features and gate structure are disposed over the same fins. For example, the device having gate structure **203**, S/D features below S/D contacts **211** and **213** is considered a single pFET even when the device spans across fin **204** and fin **204**A. The same applies to FETs that have gate structures and S/D features spanning across more than two fins.

The S/D features may be in-situ doped during the epitaxy process by introducing doping species including: p-type dopants, such as boron or BF2; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. If the S/D features are not in-situ doped, an implantation process (i.e., a junction implant process) is performed to introduce the corresponding dopant into the S/D features. In an exemplary embodiment, S/D features of a pFET include SiGeB, while S/D features an nFET include SiP. One or more annealing processes may be performed thereafter to activate the S/D features. Suitable annealing processes include rapid thermal annealing (RTA), laser annealing processes, other suitable annealing technique or a combination thereof.

In some instances, a dielectric material layer, sometimes referred to as interlayer dielectric (ILD), is to be deposited over the substrate, filling the space between the S/D features and the space between gate structures. The ILD layer is to undergo a chemical mechanical polishing process for planarization.

S/D contacts are formed over S/D features. An anisotropic etching process is used to form an opening through the ILD layer and S/D features. Then, the opening is filled with conductive material. In some instances, before conductive material is filled in the opening, silicide may be formed in the opening to reduce contact resistance. Silicide can be formed by reacting silicon with a metal, such as titanium, tantalum, nickel or cobalt. In some examples, the silicide may be formed by a process referred to as self-aligned silicide or salicide. The salicide process includes depositing one of the aforementioned metals, annealing to cause the reaction between the metal and silicon, and removing unreacted metal materials. To prevent diffusion of impurity into the conductive material, a barrier layer may be formed on the sidewall of the opening. The barrier layer may be formed of a single layer of titanium nitride (TiN) or tantalum nitride (TaN) or a multilayer such as Ti/TiN, Ta/TaN layers. In some instances, the conductive material is filled in the opening after both silicide and barrier layers are formed. Suitable conductive materials include tungsten (W), copper (Cu), aluminum (Al), and cobalt (Co).

In some embodiments, multi-fin active region **200** abuts mono-fin active region **300** along the Y direction, or the direction of the fins **202**, **202**A, **204**, **204**A, **302**, and **304**, as shown in FIG. **1**. In some implementations, although multi-fin active region **200** abuts mono-fin active region **300**, they are isolated by an isolation feature **400**A. Particularly, in FIG. **1**, no fin in multi-fin active region **200** extends continuously into mono-fin active region **300**. Isolation feature **400**A includes a dummy dielectric gate. As described above, to form a regular gate structure, a dummy gate, usually formed of poly-Si, is formed at the location where the gate structure is to be formed and the dummy gate is later removed and replaced with a high-K metal gate stack. Here, in terms of fabrication of isolation feature **400**A, a dummy gate is first formed at the location where the isolation feature **400**A is formed. However, after the dummy gate is removed, instead of replacing the dummy gate with a high-K metal gate stack, a dielectric material is used to replace the dummy gate. In some instances, isolation feature **400**A is formed of silicon oxide (SiO_2), a silicon nitride (SiN), a silicon carbide (SiC), a silicon oxynitride (SiON), other suitable materials, and/or combinations thereof.

In some embodiments, each of the S/D contacts is substantially rectangular in shape. Along the X direction, S/D contacts **210**, **211**, **212**, **213**, **214**, **215**, **310**, **311**, **312**, and **313** have lengths **X1**, **X2**, **X3**, **X4**, **X5**, **X6**, **X7**, and **X8**, respectively. In some embodiments, **X1** is substantially equal to **X2**, **X3** is substantially equal to **X4**, **X5** is substantially equal to **X6**, and **X7** is substantially equal to **X8**. In addition, along the Y direction, S/D contact **310** has a width **Y1** and S/D contact **312** has a width **Y2**. In some implementations, S/D contacts **210** and **312** are electrically connected to a Vss line, also referred to as a source node. In those implementations, X1 is more than 1.5 times of X3. Similarly, X7 is more than 1.5 times of X3. In some other embodiments, to reduce contact capacitance between S/D contact **214** and S/D contact **310**, S/D contact **310** is intentionally shortened. In some implementations where the multi-fin active region **200** includes more than two fins for both its nFET(s) and pFET(s), X3 is 1.1 to 3.0 times of X5. Put differently, in these implementations, the ratio of X3 over X5 ranges between 1.1 and 3.0. In some other implementations, X3 is 2 to 4 times of X5 if the multi-fin active region **200** includes more than two fins. In cases where multi-fin active region **200** includes two fins for both its nFET(s) and pFET(s), X**3** is about 1.1 to 2.0 times of X**5**. Put differently, the ratio of X**3** over X**5** in these cases ranges between 1.1 and 2.0. In some other implementations, X3 is 1.3 to 2.0 times of X**5** if the multi-fin active region **200** includes two fins. In some instances, Y**1** and Y**2** are substantially the same. However, in instances where the isolation feature reduces loading effect in the area near S/D contact **310**, Y**1** is about 1.1 times of Y**2**. That is, Y**1** is 10% larger than Y**2**.

Referring now to FIG. **2**, illustrated therein is a top view of semiconductor device **1100** according to aspects of the present disclosure. Semiconductor device **1100** includes fins **1202**, **1202**A, **1204**, **1204**A, **1302**, and **1304**. Fins **1202**, **1202**A, **1204**, and **1204**A are formed in a multi-fin active region **1200** while fins **1302** and **1304** are formed within a mono-fin active region **1300**. In some instances, fins **1202** and **1202**A are formed over a p-type well on semiconductor substrate **1110** and fins **1204** and **1204**A are formed over an n-type well on semiconductor substrate **1110**. In some embodiments, upon completion of the multi-fin active region **1200**, fins **1202** and **1202**A are part of at least one n-type FET (nFET) and fins **1204** and **1204**A are part of at least one p-type FET (pFET). While FIG. **2** shows that each of the nFET and pFET in multi-fin active region **1200** includes two fins, implementation with more than two fins per FET can be appreciated by people skilled in the art upon examination of the present disclosure. In cases where each FETs of multi-fin active region **1200** include two fins, the multi-fin active region can be referred to as a double-fin active region. In some embodiments, each of multi-fin active region **1200** and mono-fin active region **1300** constitutes a standard cell. In that regard, multi-fin active region **1200** can be referred to as a multi-fin standard cell and mono-fin active region **1300** can be referred to as a mono-fin standard cell.

In some embodiments, fin **1302** is formed over the same p-type well where fins **1202** and **1202**A are formed and fin **1304** is formed over the same n-type well where fins **1204** and **1204**A are formed. Upon completion of the mono-fin active region **1300**, fin **1302** is part of an nFET and fin **1304** is part of a pFET. As their names suggest, each the nFET and pFET in multi-fin active region **1200** includes more than one fin while each of the nFET and pFET in mono-fin active region **1300** includes a single fin. Semiconductor substrate **1110** and fins **1202**A, **1202**, **1204**, **1204**A, **1302** and **1304** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

As shown in FIG. **2**, in some embodiments, multi-fin active region **1200** includes gate structure **1201**, **1203**, **1205**, and **1207** and mono-fin active region **1300** includes gate structures **1305**, **1301** and **1303**. Gate structures **1201**, **1203**, **1205**, and **1207** are formed over and span across fins **1202**A, **1202**, **1204**, and **1204**A. Gate structures **1305**, **1301** and **1303** are formed over and span across fins **1302** and **1304**. In some embodiments, gate structures **1201**, **1203**, **1205**, **1301**, and **1303** contain materials and are formed in manners similar to those described above with respect to FIG. **1**. Particularly, the formation of gate structures **1201**, **1203**, **1205**, **1301**, and **1303** include formation of dummy gates and removal and replacement of those dummy gates. In some instances, gate structures **1207** and **1305** are dummy gates and are not replaced by high-K metal gate stacks. For the same reason, gate structures **1207** and **1305** may sometimes be referred to as dummy gate **1207** and dummy gate **1305**, respectively. As will be described below, gate structures **1207** and **1305** can be considered as part of isolation feature **400**B.

Source/drain (S/D) features are formed over the fins on either side of a non-floating gate structure, such as gate structure **1203**, **1205** and **1301**. As shown in FIG. **2**, for the nFET controlled by gate structure **1203** in multi-fin active region **1200**, a S/D feature is formed below S/D contact **1210** and a S/D feature is formed below S/D contact **1212**, with S/D contacts **1210** and **1212** on different sides of gate structure **1203**. In some implementations, with respect to any FET in multi-fin active region **1200**, a S/D feature and its corresponding S/D contact are formed across and span over all fins of that FET. For example, S/D contact **1210** and the S/D feature therebelow and S/D contact **1212** and the S/D feature therebelow are formed over and span across fins **1202** and **1202**A. Similarly, for the nFET controlled by gate structure **1205** in multi-fin active region **1200**, S/D contact **1212** and the S/D feature therebelow are on one side of gate structure **1205** while S/D contact **1214** and the S/D feature therebelow are on the other side of gate structure **1205**. The same applies to the pFETs in multi-fin active region **1200** and the nFET and pFET in mono-fin active region **1300**. For the pFET controlled by gate structure **1203**, S/D contact **1211** and the S/D feature therebelow are on one side of gate structure **1203** while S/D contact **1213** and S/D feature therebelow are on the other side of gate structure **1203**. For the pFET controlled by gate structure **1205**, S/D contact **1213** and the S/D feature therebelow are on one side of gate structure **1205** while S/D contact **1215** and S/D feature therebelow are on the other side of gate structure **1205**. For the nFET controlled by gate structure **1301**, S/D contact **1310** and the S/D feature therebelow are on one side of gate structure **1301** while S/D contact **1312** and S/D feature therebelow are on the other side of gate structure **1301**. Finally, for the pFET controlled by gate structure **1301**, S/D contact **1311** and the S/D feature therebelow are on one side of gate structure **1301** while S/D contact **1313** and S/D feature therebelow are on the other side of gate structure **1301**. S/D features and S/D contacts in FIG. **2** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

At least for purpose of this disclosure, an FET that includes multiple fins is still considered one FET as long as its S/D features and gate structure are disposed over the same fins. For example, the device having gate structure **1203**, S/D features below S/D contacts **1211** and **1213** is considered a single pFET even when the device spans across fin **1204** and fin **1204**A. The same applies to FETs that have gate structures and S/D features spanning across more than two fins.

In some embodiments, multi-fin active region **1200** abuts mono-fin active region **1300** along the Y direction, or the direction of the fins **1202**, **1202**A, **1204**, **1204**A, **1302**, and **1304**, as shown in FIG. **2**. In the implementations represented by FIG. **2**, although multi-fin active region **1200** abuts mono-fin active region **1300**, they are isolated by an isolation feature **400**B. Particularly, no fin in multi-fin active region **1200** extends continuously into mono-fin active region **1300**. Isolation feature **400**B includes a fin cutout region between dummy gate **1207** and dummy gate **1305**. In a broader sense, dummy gates **1207** and **1305** are considered part of isolation feature **400**B. As described above, gate structures **1207** and **1305** are dummy gates and are not replaced with any high-K metal gate stacks. In the embodiments shown in FIG. **2**, gate spacers **1120** are formed on the sidewalls of gate structures **1201**, **1203**, **1205**, **1207**, **1301**, 1303, and 1305. Gate spacers 1120 contain materials and are formed in manners similar to those described above with respect to FIG. 1.

In some embodiments, each of the S/D contacts is substantially rectangular in shape. Along the X direction, S/D contacts 1210, 1211, 1212, 1213, 1214, 1215, 1310, 1311, 1312, and 1313 have lengths X1, X2, X3, X4, X5, X6, X7, and X8, respectively. In some embodiments, X1 is substantially equal to X2, X3 is substantially equal to X4, X5 is substantially equal to X6, and X7 is substantially equal to X8. In addition, along the Y direction, S/D contact 1310 has a width Y1 and S/D contact 1312 has a width Y2. In some implementations, S/D contacts 1210 and 1312 are electrically connected to a Vss line, also referred to as a source node. In those implementations, X1 is more than 1.5 times of X3. Similarly, X7 is more than 1.5 times of X3. In some other embodiments, to reduce contact capacitance between S/D contact 1214 and S/D contact 1310, S/D contact 1310 is intentionally shortened. In some implementations where the multi-fin active region 1200 includes more than two fins for both its nFET(s) and pFET(s), X3 is 1.1 to 3.0 times of X5. Put differently, in these implementations, the ratio of X3 over X5 ranges between 1.1 and 3.0. In some other implementations, X3 is 2 to 4 times of X5 if the multi-fin active region 200 includes more than two fins. In cases where multi-fin active region 1200 includes two fins for both its nFET(s) and pFET(s), X3 is about 1.1 to 2.0 times of X5. Put differently, the ratio of X3 over X5 in these cases ranges between 1.1 and 2.0. In some other implementations, X3 is 1.3 to 2.0 times of X5 if the multi-fin active region 200 includes two fins. In some instances, Y1 and Y2 are substantially the same. However, in instances where the isolation feature reduces loading effect in the area near S/D contact 1310, Y1 is about 1.1 times of Y2. That is, Y1 is 10% larger than Y2.

Referring now to FIG. 3, illustrated therein is a top view of semiconductor device 2100 according to aspects of the present disclosure. Semiconductor device 2100 includes fins 2202, 2202A, 2204, 2204A, 2302, and 2304. Fins 2202, 2202A, 2204, and 2204A are formed in a multi-fin active region 2200 while fins 2302 and 2304 are formed within a mono-fin active region 2300. In some instances, fins 2202 and 2202A are formed over a p-type well on semiconductor substrate 2110 and fins 2204 and 2204A are formed over an n-type well on semiconductor substrate 2110. In some embodiments, upon completion of the multi-fin active region 2200, fins 2202 and 2202A are part of at least one n-type FET (nFET) and fins 2204 and 2204A are part of at least one p-type FET (pFET). While FIG. 3 shows that each of the nFET and pFET in multi-fin active region 2200 includes two fins, implementation with more than two fins per FET can be appreciated by people skilled in the art upon examination of the present disclosure. In cases where each FETs of multi-fin active region 2200 include two fins, the multi-fin active region can be referred to as a double-fin active region. In some embodiments, each of multi-fin active region 2200 and mono-fin active region 2300 constitutes a standard cell. In that regard, multi-fin active region 2200 can be referred to as a multi-fin standard cell and mono-fin active region 2300 can be referred to as a mono-fin standard cell.

In some embodiments, fin 2302 is formed over the same p-type well where fins 2202 and 2202A are formed and fin 2304 is formed over the same n-type well where fins 2204 and 2204A are formed. Upon completion of the mono-fin active region 2300, fin 2302 is part of an nFET and fin 2304 is part of a pFET. As their names suggest, each the nFET and pFET in multi-fin active region 2200 includes more than one fin while each of the nFET and pFET in mono-fin active region 2300 includes a single fin. Semiconductor substrate 2110 and fins 2202A, 2202, 2204, 2204A, 2302 and 2304 contain materials and are formed in manners similar to those described above with respect to FIG. 1.

As shown in FIG. 3, in some embodiments, multi-fin active region 2200 includes gate structure 2201, 2203, and 2205 and mono-fin active region 2300 includes gate structures 2301 and 2303. Gate structures 2201, 2203, and 2205 are formed over and span across fins 2202A, 2202, 2204, and 2204A. Gate structures 2301 and 2303 are formed over and span across fins 2302 and 2304. In some embodiments, gate structures 2201, 2203, 2205, 2301, and 2303 contain materials and are formed in manners similar to those described above with respect to FIG. 1. Particularly, the formation of gate structures 2201, 2203, 2205, 2301, and 2303 include formation of dummy gates and replacement of those dummy gates with high-k metal gate stack.

Source/drain (S/D) features are formed over the fins on either side of a non-floating gate structure, such as gate structure 2203, 2205 and 2301. As shown in FIG. 3, for the nFET controlled by gate structure 2203 in multi-fin active region 2200, a S/D feature is formed below S/D contact 2210 and a S/D feature is formed below S/D contact 2212, with S/D contacts 2210 and 2212 on different sides of gate structure 2203. In some implementations, with respect to any FET in multi-fin active region 2200, a S/D feature and its corresponding S/D contact are formed across and span over all fins of that FET. For example, S/D contact 2210 and the S/D feature therebelow and S/D contact 2212 and the S/D feature therebelow are formed over and span across fins 2202 and 2202A. Similarly, for the nFET controlled by gate structure 2205 in multi-fin active region 2200, S/D contact 2212 and the S/D feature therebelow are on one side of gate structure 2205 while S/D contact 2214 and the S/D feature therebelow are on the other side of gate structure 2205. The same applies to the pFETs in multi-fin active region 2200 and the nFET and pFET in mono-fin active region 2300. For the pFET controlled by gate structure 2203, S/D contact 2211 and the S/D feature therebelow are on one side of gate structure 2203 while S/D contact 2213 and S/D feature therebelow are on the other side of gate structure 2203. For the pFET controlled by gate structure 2205, S/D contact 2213 and the S/D feature therebelow are on one side of gate structure 2205 while S/D contact 2215 and S/D feature therebelow are on the other side of gate structure 2205. For the nFET controlled by gate structure 2301, S/D contact 2310 and the S/D feature therebelow are on one side of gate structure 2301 while S/D contact 2312 and S/D feature therebelow are on the other side of gate structure 2301. Finally, for the pFET controlled by gate structure 2301, S/D contact 2311 and the S/D feature therebelow are on one side of gate structure 2301 while S/D contact 2313 and S/D feature therebelow are on the other side of gate structure 2301. S/D features and S/D contacts in FIG. 3 contain materials and are formed in manners similar to those described above with respect to FIG. 1.

At least for purpose of this disclosure, an FET that includes multiple fins is still considered one FET as long as its S/D features and gate structure are disposed over the same fins. For example, the device having gate structure 2203, S/D features below S/D contacts 2211 and 2213 is considered a single pFET even when the device spans across fin 2204 and fin 2204A. The same applies to FETs that have gate structures and S/D features spanning across more than two fins.

In some embodiments, multi-fin active region **2200** abuts mono-fin active region **2300** along the Y direction, or the direction of the fins **2202**, **2202**A, **2204**, **2204**A, **2302**, and **2304**, as shown in FIG. **3**. In the implementations represented by FIG. **3**, although multi-fin active region **2200** abuts mono-fin active region **2300**, they are isolated by an isolation feature **400**C. Particularly, no fin in multi-fin active region **2200** extends continuously into mono-fin active region **2300**. Isolation feature **400**C includes a gate-free region. That is, no gate structure, including any dummy gate, dummy dielectric gate, is formed in the area where isolation feature **400**C is located. In these implementations, isolation feature includes an ILD layer. The ILD layer contains materials and is formed in manners similar to those described above with respective to FIG. **1**. In the embodiments shown in FIG. **3**, gate spacers **2120** are formed on the sidewalls of gate structures **2201**, **2203**, **2205**, **2301** and **2303**. Gate spacers **2120** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

In some embodiments, each of the S/D contacts is substantially rectangular in shape. Along the X direction, S/D contacts **2210**, **2211**, **2212**, **2213**, **2214**, **2215**, **2310**, **2311**, **2312**, and **2313** have lengths **X1**, **X2**, **X3**, **X4**, **X5**, **X6**, **X7**, and **X8**, respectively. In some embodiments, **X1** is substantially equal to **X2**, **X3** is substantially equal to **X4**, **X5** is substantially equal to **X6**, and **X7** is substantially equal to **X8**. In addition, along the Y direction, S/D contact **2310** has a width **Y1** and S/D contact **2312** has a width **Y2**. In some implementations, S/D contacts **2210** and **2312** are electrically connected to a Vss line, also referred to as a source node. In those implementations, **X1** is more than 1.5 times of **X3**. Similarly, **X7** is more than 1.5 times of **X3**. In some other embodiments, to reduce contact capacitance between S/D contact **2214** and S/D contact **2310**, S/D contact **2310** is intentionally shortened. In some implementations where the multi-fin active region **2200** includes more than two fins for both its nFET(s) and pFET(s), **X3** is 1.1 to 3.0 times of **X5**. Put differently, in these implementations, the ratio of **X3** over **X5** ranges between 1.1 and 3.0. In some other implementations, **X3** is 2 to 4 times of **X5** if the multi-fin active region **200** includes more than two fins. In cases where multi-fin active region **2200** includes two fins for both its nFET(s) and pFET(s), **X3** is about 1.1 to 2.0 times of **X5**. Put differently, the ratio of **X3** over **X5** in these cases ranges between 1.1 and 2.0. In some other implementations, **X3** is 1.3 to 2.0 times of **X5** if the multi-fin active region **200** includes two fins. In some instances, **Y1** and **Y2** are substantially the same. However, in instances where the isolation feature reduces loading effect in the area near S/D contact **2310**, **Y1** is about 1.1 times of **Y2**. That is, **Y1** is 10% larger than **Y2**.

Referring now to FIG. **4**, illustrated therein is a top view of semiconductor device **3100** according to aspects of the present disclosure. Semiconductor device **3100** includes fins **3202**, **3202**A, **3204**, and **3204**A. Fins **3202**A and **3204**A extend from multi-fin active region **3200** into mono-fin active region **3300** while fins **3202** and **3204** do not. In some instances, fins **3202** and **3202**A are formed over a p-type well on semiconductor substrate **3110** and fins **3204** and **3204**A are formed over an n-type well on semiconductor substrate **3110**. In some embodiments, upon completion of the multi-fin active region **3200**, fins **3202** and **3202**A are part of at least one n-type FET (nFET) and fins **3204** and **3204**A are part of at least one p-type FET (pFET) in multi-fin active region. While FIG. **4** shows that each of the nFET and pFET in multi-fin active region **3200** includes two fins, implementation with more than two fins per FET can be appreciated by people skilled in the art upon examination of the present disclosure. In cases where each FETs of multi-fin active region **3200** include two fins, the multi-fin active region can be referred to as a double-fin active region. In some embodiments, each of multi-fin active region **3200** and mono-fin active region **3300** constitutes a standard cell. In that regard, multi-fin active region **3200** can be referred to as a multi-fin standard cell and mono-fin active region **3300** can be referred to as a mono-fin standard cell.

In the embodiments represented by FIG. **4**, the portion of fin **3202**A in mono-fin active region **3300** is formed over the same p-type well where fins **3202** and **3202**A are formed and the portion of the fin **3204**A in mono-fin active region **3300** is formed over the same n-type well where fins **3204** and **3204**A are formed. Upon completion of the mono-fin active region **3300**, the portion of fin **3202**A in mono-fin region **3300** is part of at least one nFET in mono-fin active region and the portion of fin **3204**A in mono-fin region **3300** is part of at least one pFET in mono-fin active region. As their names suggest, each the nFET and pFET in multi-fin active region **3200** includes more than one fin while each of the nFET and pFET in mono-fin active region **3300** includes a single fin. Semiconductor substrate **3110** and fins **3202**A, **3202**, **3204**, and **3204**A contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

As shown in FIG. **4**, in some embodiments, multi-fin active region **3200** includes gate structure **3201**, **3203**, and **3205** and mono-fin active region **3300** includes gate structures **3301** and **3303**. Gate structures **3201**, **3203**, and **3205** are formed over and span across fins **3202**A, **3202**, **3204**, and **3204**A. Gate structures **3301** and **3303** are formed over and span across the portions of fins **3202**A and **3204**A in mono-fin active region **3300**. In the embodiment shown in FIG. **4**, semiconductor device **3100** also include a gate structure **3410** and a gate structure **3420**, both of which are disposed parallel to gate structures **3201**, **3203**, **3205**, **3301**, and **3303** along the X direction. Gate structures **3410** and **3420** are positioned between multi-fin active region **3200** and mono-fin active region **3300** and serve as parts of an isolation feature **400**D to isolate multi-fin active region **3200** from mono-fin active region **3300**. In some implementations, gate structure **3410** is formed over fins **3202** and **3202**A, with fin **3202** only extending about halfway thereunder. Gate structure **3420** is formed over fins **3204** and **3204**A, with fin **3204** extending about halfway thereunder. In some embodiments, gate structures **3410** and **3420** are aligned in the X direction but are separate from one another. In some embodiments, gate structures **3201**, **3203**, **3205**, **3301**, **3303**, **3410**, and **3420** contain materials and are formed in manners similar to those described above with respect to FIG. **1**. Particularly, the formation of gate structures **3201**, **3203**, **3205**, **3301**, **3303**, **3410** and **3420** include formation of dummy gates and replacement of those dummy gates with high-k metal gate stack. Compared to gate structures **3201**, **3203**, **3205**, **3301**, and **3303**, formation of gate structures **3410** and **3420** requires additional process steps. For example, to form gate structures **3410** and **3420**, a continuous dummy gate is first formed over the location where gate structures **3410** and **3420** would be positioned. After the dummy gate is removed and before high-K metal gate stack is formed in place of the removed dummy gate, a dielectric feature is formed in the middle (where gate structures **3410** and **3420** are separated) as a separation. Thereafter the high-K metal gate stack is formed on either side of the dielectric feature, thus forming separate gate structures **3410** and **3420**. Gate structures **3410** and **3420** may be formed by other processes. For instance, after the dummy gate is replaced with high-K metal gate stack, a middle portion of the high-K metal gate stack is removed by etching to separate gate structures **3410** and **3420**. In the embodiments shown in FIG. **4**, gate spacers **3120** are formed on the sidewalls of gate structures **3201**, **3203**, **3205**, **3301**, **3303**, **3410**, and **3420**. Gate spacers **3120** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

Source/drain (S/D) features are formed over the fins on either side of a non-floating gate structure, such as gate structure **3203**, **3205** and **3301**. As shown in FIG. **4**, for the nFET controlled by gate structure **3203** in multi-fin active region **3200**, a S/D feature is formed below S/D contact **3210** and a S/D feature is formed below S/D contact **3212**, with S/D contacts **3210** and **3212** on different sides of gate structure **3203**. In some implementations, with respect to any FET in multi-fin active region **3200**, an S/D feature and its corresponding S/D contact are formed across and span over all fins of that FET. For example, S/D contact **3210** and the S/D feature therebelow and S/D contact **3212** and the S/D feature therebelow are formed over and span across fins **3202** and **3202**A. Similarly, for the nFET controlled by gate structure **3205** in multi-fin active region **3200**, S/D contact **3212** and the S/D feature therebelow are on one side of gate structure **3205** while S/D contact **3214** and the S/D feature therebelow are on the other side of gate structure **3205**. The same applies to the pFETs in multi-fin active region **3200** and the nFET and pFET in mono-fin active region **3300**. For the pFET controlled by gate structure **3203**, S/D contact **3211** and the S/D feature therebelow are on one side of gate structure **3203** while S/D contact **3213** and S/D feature therebelow are on the other side of gate structure **3203**. For the pFET controlled by gate structure **3205**, S/D contact **3213** and the S/D feature therebelow are on one side of gate structure **3205** while S/D contact **215** and S/D feature therebelow are on the other side of gate structure **3205**. For the nFET controlled by gate structure **3301**, S/D contact **3310** and the S/D feature therebelow are on one side of gate structure **3301** while S/D contact **3312** and S/D feature therebelow are on the other side of gate structure **3301**. Finally, for the pFET controlled by gate structure **3301**, S/D contact **3311** and the S/D feature therebelow are on one side of gate structure **3301** while S/D contact **3313** and S/D feature therebelow are on the other side of gate structure **3301**. S/D features and S/D contacts in FIG. **4** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

At least for purpose of this disclosure, an FET that includes multiple fins is still considered one FET as long as its S/D features and gate structure are disposed over the same fins. For example, the device having gate structure **3203**, S/D features below S/D contacts **3211** and **3213** is considered a single pFET even when the device spans across fin **3204** and fin **3204**A. The same applies to FETs that have gate structures and S/D features spanning across more than two fins.

In some embodiments, multi-fin active region **3200** abuts mono-fin active region **3300** along the Y direction, or the direction of the fins **3202**, **3202**A, **3204**, and **3204**A, as shown in FIG. **4**. In the implementations represented by FIG. **4**, although multi-fin active region **3200** abuts mono-fin active region **3200**, they are isolated by isolation feature **400**D. In some embodiments, isolation feature **400**D includes an isolation nFET and an isolation pFET. The isolation nFET includes fin **3202**A formed over a p-type well on semiconductor substrate **3110**, gate structure **3410**, S/D contact **3214** and the S/D feature therebelow, and S/D contact **3310** and S/D feature therebelow. By way of an interconnect via **3415**, gate structure **3410** is electrically connected to a Vss line, which is usually referred to as a low voltage line or a source node, and the isolation nFET formed over gate structure **3410** is constantly turned off, shutting down the channels in fin **3202**A across gate structure **3410**. The isolation pFET includes fin **3204**A formed over an n-type well on semiconductor substrate **3110**, gate structure **3420**, S/D contact **3215** and the S/D feature therebelow, and S/D contact **3311** and S/D feature therebelow. By way of an interconnect via **3425**, gate structure **3420** is electrically connected to a Vdd line, which is usually the high voltage line, and the isolation pFET formed over gate structure **3420** is constantly turned off, shutting down the channels in fin **3204**A across gate structure **3420**. As both isolation nFET and pFET are turned off, multi-fin active region **3200** is effectively isolated from mono-fin active region **3300** by isolation feature **400**D.

In some embodiments, each of the S/D contacts is substantially rectangular in shape. Along the X direction, S/D contacts **3210**, **3211**, **3212**, **3213**, **3214**, **3215**, **3310**, **3311**, **3312**, and **3313** have lengths X**1**, X**2**, X**3**, X**4**, X**5**, X**6**, X**7**, and X**8**, respectively. In some embodiments, X**1** is substantially equal to X**2**, X**3** is substantially equal to X**4**, X**5** is substantially equal to X**6**, and X**7** is substantially equal to X**8**. In addition, along the Y direction, S/D contact **3310** has a width Y**1** and S/D contact **3312** has a width Y**2**. In some implementations, S/D contacts **3210** and **3312** are electrically connected to a Vss line, also referred to as a source node. In those implementations, X**1** is more than 1.5 times of X**3**. Similarly, X**7** is more than 1.5 times of X**3**. In some other embodiments, to reduce contact capacitance between S/D contact **3214** and S/D contact **3310**, S/D contact **3310** is intentionally shortened. In some implementations where the multi-fin active region **3200** includes more than two fins for both its nFET(s) and pFET(s), X**3** is 1.1 to 3.0 times of X**5**. Put differently, in these implementations, the ratio of X**3** over X**5** ranges between 1.1 and 3.0. In some other implementations, X**3** is 2 to 4 times of X**5** if the multi-fin active region **200** includes more than two fins. In cases where multi-fin active region **3200** includes two fins for both its nFET(s) and pFET(s), X**3** is about 1.1 to 2.0 times of X**5**. Put differently, the ratio of X**3** over X**5** in these cases ranges between 1.1 and 2.0. In some other implementations, X**3** is 1.3 to 2.0 times of X**5** if the multi-fin active region **200** includes two fins. In some instances, Y**1** and Y**2** are substantially the same. However, in instances where the isolation feature reduces loading effect in the area near S/D contact **3310**, Y**1** is about 1.1 times of Y**2**. That is, Y**1** is 10% larger than Y**2**.

Referring now to FIG. **5**, illustrated therein is a top view of semiconductor device **4100** according to aspects of the present disclosure. Semiconductor device **4100** includes fins **4202**, **4202**A, **4204**, and **4204**A. Different from the embodiment shown in FIG. **4**, fins **4202** and **4204**, not fins **4202**A and **4204**A, extend from multi-fin active region **4200** into mono-fin active region **4300** while fins **4202**A and **4204**A do not. In some instances, fins **4202** and **4202**A are formed over a p-type well on semiconductor substrate **4110** and fins **4204** and **4204**A are formed over an n-type well on semiconductor substrate **4110**. In some embodiments, upon completion of the multi-fin active region **4200**, fins **4202** and **4202**A are part of at least one n-type FET (nFET) and fins **4204** and **4204**A are part of at least one p-type FET (pFET) in multi-fin active region. While FIG. **5** shows that each of the nFET and pFET in multi-fin active region **4200** includes two fins, implementation with more than two fins per FET can be appreciated by people skilled in the art upon examination of the present disclosure. In cases where each FETs of multi-fin active region **4200** include two fins, the multi-fin active region can be referred to as a double-fin active region. In some embodiments, each of multi-fin active region **4200** and mono-fin active region **4300** constitutes a standard cell. In that regard, multi-fin active region **4200** can be referred to as a multi-fin standard cell and mono-fin active region **4300** can be referred to as a mono-fin standard cell.

In the embodiments represented by FIG. **5**, the portion of fin **4202** in mono-fin active region **4300** is formed over the same p-type well where fins **4202** and **4202**A are formed and the portion of the fin **4204** in mono-fin active region **4300** is formed over the same n-type well where fins **4204** and **4204**A are formed. Upon completion of the mono-fin active region **4300**, the portion of fin **4202** in mono-fin region **4300** is part of at least one nFET in mono-fin active region and the portion of fin **4204** in mono-fin region **4300** is part of at least one pFET in mono-fin active region. As their names suggest, each the nFET and pFET in multi-fin active region **4200** includes more than one fin while each of the nFET and pFET in mono-fin active region **4300** includes a single fin. Semiconductor substrate **4110** and fins **4202**A, **4202**, **4204**, and **4204**A contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

As shown in FIG. **5**, in some embodiments, multi-fin active region **4200** includes gate structure **4201**, **4203**, and **4205** and mono-fin active region **4300** includes gate structures **4301** and **4303**. Gate structures **4201**, **4203**, and **4205** are formed over and span across fins **4202**A, **4202**, **4204**, and **4204**A. Gate structures **4301** and **4303** are formed over and span across the portions of fins **4202** and **4204** in mono-fin active region **4300**. In the embodiment shown in FIG. **5**, semiconductor device **4100** also include a gate structure **4410** and a gate structure **4420**, both of which are disposed parallel to gate structures **4201**, **4203**, **4205**, **4301**, and **4303** along the X direction. Gate structures **4410** and **4420** are positioned between multi-fin active region **4200** and mono-fin active region **4300** and serve as parts of an isolation feature **400**E to isolate multi-fin active region **4200** from multi-fin active region. In some implementations, gate structure **4410** is formed over fins **4202** and **4202**A, with fin **4202**A only extending about halfway thereunder. Gate structure **4420** is formed over fins **4204** and **4204**A, with fin **4204**A extending about halfway thereunder. In some embodiments, gate structures **4410** and **4420** are aligned in the X direction but are separate from one another. In some embodiments, gate structures **4201**, **4203**, **4205**, **4301**, **4303**, **4410**, and **4420** contain materials and are formed in manners similar to those described above with respect to FIG. **1**. Particularly, the formation of gate structures **4201**, **4203**, **4205**, **4301**, **4303**, **4410** and **4420** include formation of dummy gates and replacement of those dummy gates with high-k metal gate stack. Compared to gate structures **4201**, **4203**, **4205**, **4301**, and **4303**, formation of gate structures **4410** and **4420** requires additional process steps. For example, to form gate structures **4410** and **4420**, a continuous dummy gate is first formed over the location where gate structures **4410** and **4420** would be positioned. After the dummy gate is removed and before high-K metal gate stack is formed in place of the removed dummy gate, a dielectric feature is formed in the middle (where gate structures **4410** and **4420** would be separated) as a separation. Thereafter the high-K metal gate stack is formed on either side of the dielectric feature, thus forming separate gate structures **4410** and **4420**. Gate structures **4410** and **4420** may be formed by other processes. For instance, after the dummy gate is replaced with high-K metal gate stack, a middle portion of the high-K metal gate stack is removed by etching to separate gate structures **4410** and **4420**. In the embodiments shown in FIG. **5**, gate spacers **4120** are formed on the sidewalls of gate structures **4201**, **4203**, **4205**, **4301**, **4303**, **4410**, and **4420**. Gate spacers **4120** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

Source/drain (S/D) features are formed over the fins on either side of a non-floating gate structure, such as gate structure **4203**, **4205** and **4301**. As shown in FIG. **5**, for the nFET controlled by gate structure **4203** in multi-fin active region **4200**, an S/D feature is formed below S/D contact **4210** and a S/D feature is formed below S/D contact **4212**, with S/D contacts **4210** and **4212** on different sides of gate structure **4203**. In some implementations, with respect to any FET in multi-fin active region **4200**, an S/D feature and its corresponding S/D contact are formed across and span over all fins of that FET. For example, S/D contact **4210** and the S/D feature therebelow and S/D contact **4212** and the S/D feature therebelow are formed over and span across fins **4202** and **4202**A. Similarly, for the nFET controlled by gate structure **4205** in multi-fin active region **4200**, S/D contact **4212** and the S/D feature therebelow are on one side of gate structure **4205** while S/D contact **4214** and the S/D feature therebelow are on the other side of gate structure **4205**. The same applies to the pFETs in multi-fin active region **4200** and the nFET and pFET in mono-fin active region **4300**. For the pFET controlled by gate structure **4203**, S/D contact **4211** and the S/D feature therebelow are on one side of gate structure **4203** while S/D contact **4213** and S/D feature therebelow are on the other side of gate structure **4203**. For the pFET controlled by gate structure **4205**, S/D contact **4213** and the S/D feature therebelow are on one side of gate structure **4205** while S/D contact **4215** and S/D feature therebelow are on the other side of gate structure **4205**. For the nFET controlled by gate structure **4301**, S/D contact **4310** and the S/D feature therebelow are on one side of gate structure **4301** while S/D contact **4312** and S/D feature therebelow are on the other side of gate structure **4301**. Finally, for the pFET controlled by gate structure **4301**, S/D contact **4311** and the S/D feature therebelow are on one side of gate structure **4301** while S/D contact **4313** and S/D feature therebelow are on the other side of gate structure **4301**. S/D features and S/D contacts in FIG. **5** contain materials and are formed in manners similar to those described above with respect to FIG. **1**.

At least for purpose of this disclosure, an FET that includes multiple fins is still considered one FET as long as its S/D features and gate structure are disposed over the same fins. For example, the device having gate structure **4203**, S/D features below S/D contacts **4211** and **4213** is considered a single pFET even when the device spans across fin **4204** and fin **4204**A. The same applies to FETs that have gate structures and S/D features spanning across more than two fins.

In some embodiments, multi-fin active region **4200** abuts mono-fin active region **4300** along the Y direction, or the direction of the fins **4202**, **4202**A, **4204**, and **4204**A, as shown in FIG. **5**. In the implementations represented by FIG. **5**, although multi-fin active region **4200** abuts mono-fin active region **4300**, they are isolated by isolation feature **400**E. In some embodiments, isolation feature **400**E includes an isolation nFET and an isolation pFET. The isolation nFET includes fin **4202** formed over a p-type well on semiconductor substrate **4110**, gate structure **4410**, S/D contact **4214** and the S/D feature therebelow, and S/D contact **4310** and S/D feature therebelow. By way of an interconnect via **4415**, gate structure **4410** is electrically connected to a Vss line, which is usually referred to as a low voltage line or a source node, and the isolation nFET formed over gate structure **4410** is constantly turned off, shutting down the channels in fin **4202** across gate structure **4410**. The isolation pFET includes fin **4204** formed over an n-type well on semiconductor substrate **4110**, gate structure **4420**, S/D contact **4215** and the S/D feature therebelow, and S/D contact **4311** and S/D feature therebelow. By way of an interconnect via **4425**, gate structure **4420** is electrically connected to a Vdd line, which is usually the high voltage line, and the isolation pFET formed over gate structure **4420** is constantly turned off, shutting down the channels in fin **4204** across gate structure **4420**. As both isolation nFET and pFET are turned off, multi-fin active region **4200** is effectively isolated from mono-fin active region **4300** by isolation feature **400**E.

In some embodiments, each of the S/D contacts is substantially rectangular in shape. Along the X direction, S/D contacts **4210**, **4211**, **4212**, **4213**, **4214**, **4215**, **4310**, **4311**, **4312**, and **4313** have lengths X**1**, X**2**, X**3**, X**4**, X**5**, X**6**, X**7**, and X**8**, respectively. In some embodiments, X**1** is substantially equal to X**2**, X**3** is substantially equal to X**4**, X**5** is substantially equal to X**6**, and X**7** is substantially equal to X**8**. In addition, along the Y direction, S/D contact **4310** has a width Y**1** and S/D contact **4312** has a width Y**2**. In some implementations, S/D contacts **4210** and **4312** are electrically connected to a Vss line, also referred to as a source node. In those implementations, X**1** is more than 1.5 times of X**3**. Similarly, X**7** is more than 1.5 times of X**3**. In some other embodiments, to reduce contact capacitance between S/D contact **4214** and S/D contact **4310**, S/D contact **4310** is intentionally shortened. In some implementations where the multi-fin active region **4200** includes more than two fins for both its nFET(s) and pFET(s), X**3** is 1.1 to 3.0 times of X**5**. Put differently, in these implementations, the ratio of X**3** over X**5** ranges between 1.1 and 3.0. In some other implementations, X**3** is 2 to 4 times of X**5** if the multi-fin active region **200** includes more than two fins. In cases where multi-fin active region **4200** includes two fins for both its nFET(s) and pFET(s), X**3** is about 1.1 to 2.0 times of X**5**. Put differently, the ratio of X**3** over X**5** in these cases ranges between 1.1 and 2.0. In some other implementations, X**3** is 1.3 to 2.0 times of X**5** if the multi-fin active region **4200** includes two fins. In some instances, Y**1** and Y**2** are substantially the same. However, in instances where the isolation feature reduces loading effect in the area near S/D contact **4310**, Y**1** is about 1.1 times of Y**2**. That is, Y**1** is 10% larger than Y**2**.

Each of multi-fin active regions **200**, **1200**, **2200**, **3200**, **4200** and mono-fin active regions **300**, **1300**, **2300**, **3300**, **4300** can be viewed as a standard cell. That is, semiconductor device **100** (or **1100**, **2100**, **3100**, **4100**) is considered a column (along the X-direction) of two standard cells. Persons skilled in the art, upon examination of the present disclosure, would appreciate that the present disclosure contemplates not only a column of few standard cells but an array of a large number of standard cells. For illustration purposes, FIGS. **1-5** show two standard cells to demonstrate advantages of the semiconductor device **100** (or **1100**, **2100**, **3100**, **4100**) according to aspects of the present disclosure.

Referring now to FIG. **6**, shown therein is diagrammatic top view of two columns of standard cells **510** and **520** in a semiconductor cell array **500**, according to aspects of the present disclosure. For illustration purposes, FIG. **6** shows that the first column of standard cells **510** (hereinafter referred to as "first column") includes standard cells **511**, **512**, **513**, **514**, and **515** and second column of standard cells ("second column") includes standard cells **521**, **522**, **523**, **524**, and **525**. The standard cells **511-515** span across an n-type well **610** and a p-type well **620** and the standard cells **521-525** span across the p-type well **620** and another n-type well **630**. Arranged in this manner, standard cells **511-515** of the first column **510** share the same p-type well **620** with standard cells **521-525** of the second column **520**. Furthermore, in these implementations, the nFETs of standard cells **511-515** are adjacent to nFETs of standard cells **521-525**. The semiconductor cell array **500** can also be rearranged so that the first column **510** and the second column **520** share the same n-type well. For example, the first column **510** can have its pFETs on the left-hand side and its nFET on the right-hand side while the second column **520** can have its pFET on the right-hand side and its nFET on the left-hand side. That way, the pFETs of the first column **510** can share the same n-type well with the pFETs of the second column **520**. In some instances, the semiconductor cell array **500** is scalable by having more rows of standard cells abutting the first column **510** or the second column **520** or by adding more standard cells to first column **510** and second column **520**. In those instances, a column of standard cells adjacent to the first column **510** can have pFETs that share the same n-type well **610** with standard cells **511-515** in first column **510**. Similarly, a column of standard cells adjacent to the second column **520** can have pFETs that share the same n-type well **630** with standard cells **521-525** in second column **520**.

Each of the standard cells **511-15** and **521-525** can consist of a multi-fin active region to function as a multi-fin device or of a mono-fin active region to function as a mono-fin device. Sometimes a standard cell comprised of a multi-fin active region can be referred to a multi-fin standard cell. Similarly, a standard cell comprised of a mono-fin active region can be referred to a mono-fin standard cell. For example, with respect to first column **510**, as illustrated in FIG. **6**, standard cells **512** and **514** are multi-fin devices and standard cells **511**, **513** and **515** are mono-fin devices. With respect to the second column **520**, standard cells **521**, **524** and **525** are multi-fin devices while standard cell **522** and **523** are mono-fin devices. In terms of circuit design, standard cells **512**, **514**, **521**, **524**, and **525** are multi-fin standard cells and standard cells **511**, **513**, **515**, **522**, and **523** are mono-fin standard cells. In some implementations, the first column **510** and the second column **520** span along the same length along the length-wise direction of the fins that are disposed over the first column **510** and the second column **520**. In those implementations, the first column **510** and the second column **520** are substantially rectangular in shape and have the same length along the length-wise direction of the fins. In some instances, the length of the first and second rows **510** and **520** along the Y direction are proportional to the number of gate structure (including dielectric dummy gates). That is to say, when the length of the first column **510** is substantially equal to the length of the second column **520**, the number of gate electrodes in the first column **510** and the second column **520** are substantially similar.

Depending on the design of the semiconductor device **100** or semiconductor cell array **500**, mono-fin active region **300** (or a mono-fin standard cell) and multi-fin active region **200** (or a multi-fin standard cell) can include a variety of devices. In circuit design, while fast switching speed and low leakage current are almost always preferred, sometimes they cannot be achieved at the same time and tradeoffs have to be made.

In those situations, if speed is more important than leakage in certain node or application, a multi-fin standard cell is preferred over a mono-fin standard cell. However, if leakage is more important than speed in another node or application, then a mono-fin standard cell may be more preferable than a multi-fin standard cell. Conventionally, if a circuit demands both mono-fin devices and multi-fin devices, they are fabricated in different colonies on a substrate and are connected via interconnect structures. The embodiments in the present disclosure allow multi-fin standard cells and mono-fin standard cells to be placed next to one another to meet the design requirements in speed and leakages, thus eliminating the need for complex interconnect structures.

Thus, the present disclosure provides examples of a semiconductor device and a semiconductor cell array. In some embodiments, a semiconductor device includes a multi-fin active region, a mono-fin active region, and an isolation feature between the multi-fin active region and the mono-fin active region. The multi-fin active region includes a first plurality of fins, a second plurality of fins parallel to the first plurality of fins, a first n-type field effect transistor (FET), a first p-type FET. The first n-type FET includes a first gate structure disposed over the first plurality of fins and first source/drain (S/D) contacts on either side of the first gate structure. Each of the S/D contacts is disposed over the first plurality of fins. The first p-type FET includes the first gate structure disposed over the second plurality of fins and second S/D contacts on either side of the first gate structure. Each of the S/D contacts is disposed over the second plurality of fins. The mono-fin active region abuts the multi-fin active region. The mono-fin active region includes a first fin, a second fin different from the first fin, a second n-type FET, and a second p-type FET. The second n-type FET includes a second gate structure disposed over the first fin and third S/D contacts on either side of the second gate structure. Each of the third S/D contacts is disposed over the first fin. The second p-type FET includes the second gate structure disposed over the second fin and fourth S/D contacts on either side of the second gate structure. Each of the fourth S/D contacts is disposed over the second fin. The isolation feature is parallel to the first and second gate structures.

In some such embodiments, the first fin of the semiconductor device is aligned with one of the first plurality of fins and the second fin is aligned with one of the second plurality of fins. In some implementations, the isolation feature of the semiconductor device includes a dummy dielectric gate. In some other implementations, the isolation feature of the semiconductor device includes a fin cutout region. In still some other implementations, the isolation feature of the semiconductor device includes an isolation p-type FET and an isolation n-type FET. The isolation p-Type FET includes a gate electrically connected to a Vdd line and the isolation n-type FET includes a gate electrically connected to a Vss line. In some embodiments, the first and second S/D contacts of the semiconductor device includes a first group of S/D contacts away from the mono-fin active region and a second group of S/D contacts adjacent to the mono-fin active region. Each of the first group of S/D contacts has a first length parallel to the first gate structure and each of the second group of S/D contacts has a second length parallel to the first gate structure. The first length is greater than the second length. In some such embodiments, the third and fourth S/D contacts of the semiconductor device include a third group of S/D contacts adjacent to the multi-fin active region and a fourth group of S/D contacts away from the multi-fin active region. Each of the third group of S/D contacts has a third length parallel to the second gate structure and each of the fourth group of S/D contacts has a fourth length parallel to the second gate structure. The third length is smaller than the second length. In some such embodiments, the first length is at least 1.5 times of the second length, the first plurality of fins comprises more than two fins, the second plurality of fins comprises more than two fins, and a ratio of the second length over the third length ranges between 2 and 4. In some such embodiments, each of the third group of S/D contacts has a first width parallel to the first fin and each of the fourth group of S/D contacts has a second width parallel to the first fin. The first width is larger than the second width.

In further embodiments, a semiconductor cell array includes a first column of standard cells and a second column of standard cells. Each of the first column of standard cells includes a p-type field effect transistor (FET) and an n-type FET. Each of the second column of standard cells includes a p-type FET and an n-type FET. The n-type FETs in the first column of standard cells share a same p-type well with the n-type FETs in the second column of standard cells. The first column of standard cells includes a multi-fin standard cell and a mono-fin standard cell. The multi-fin standard cell is adjacent to the mono-fin standard cell. The multi-fin standard cell is isolated from the mono-fin standard cell by an isolation feature. In some such embodiments, the multi-fin standard cell includes a first gate structure and first source/drain (S/D) contacts adjacent to mono-fin standard cell. The first gate structure and the first S/D contacts are disposed over a plurality of fins. The mono-fin standard cell includes a second gate structure parallel to the first gate structure and second source/drain (S/D) contacts adjacent to the multi-fin standard cell. The second gate structure and the second S/D contacts are disposed over a fin. In some instances, each of the first S/D contacts includes a first length parallel to the first gate structure and each of the second S/D contacts includes a second length parallel to the second gate structure. In some other instances, a ratio of the first length over the second length ranges between 1.1 and 3.0. In some other embodiments, the multi-fin standard cell abuts the mono-fin standard cells such that the fin is aligned with one of the plurality of fins. In some instances, each of the first column of standard cells and the second column of standard cells is rectangular in shape and a total number of gate structures in the first column of standard cells is identical to a total number of gate structures in the second column of standard cells.

In yet further embodiments, a semiconductor device includes a double-fin active region, a mono-fin active region abutting the double-fin active region, and an isolation feature between the double-fin active region and the mono-fin active region. The double-fin active region includes a first pair of fins, a second pair of fins parallel to the first pair of fins, a first n-type field effect transistor (FET) including a first gate structure disposed over the first pair of fins and first source/drain (S/D) contacts on either side of the first gate structure, each of the first S/D contacts being disposed over the first pair of fins, a first p-type FET, including the first gate structure disposed over the second pair of fins and second source/drain (S/D) contacts on either side of the first gate structure, each of the second S/D contacts being disposed over the second pair of fins. The mono-fin active region includes a first fin, a second fin different from the first fin, a second n-type FET including a second gate structure disposed over the first fin and third S/D contacts on either side of the second gate structure, each of the third S/D contacts being disposed over the first fin, a second p-type FET including the second gate structure disposed over the second fin and fourth S/D contacts on either side of the second gate structure, each of the fourth S/D contacts being disposed over the second fin; and an isolation feature between the double-fin active region and the mono-fin active region, the isolation feature being parallel to the first and second gate structures. In some such embodiments, the first fin is aligned with one of the first pair of fins and the second fin is aligned with one of the second pair of fins. In some implementations, the isolation feature comprises a dummy dielectric gate. In some other embodiments, the isolation feature includes a fin cutout region. In some instances, the isolation feature includes an isolation p-type FET and an isolation n-type FET. In those instances, the isolation p-Type FET includes a gate electrically connected to a Vdd line and the isolation n-type FET includes a gate electrically connected to a Vss line.

In some embodiments, the first and second S/D contacts of the semiconductor device includes a first group of S/D contacts away from the mono-fin active region and a second group of S/D contacts adjacent to the mono-fin active region. Each of the first group of S/D contacts has a first length parallel to the first gate structure. Each of the second group of S/D contacts has a second length parallel to the first gate structure. The first length is greater than the second length. The third and fourth S/D contacts include a third group of S/D contacts adjacent to the double-fin active region and a fourth group of S/D contacts away from the double-fin active region. Each of the third group of S/D contacts has a third length parallel to the second gate structure and each of the fourth group of S/D contacts has a fourth length parallel to the second gate structure. The second length is greater than the third length. In some implementations, a ratio of the second length over the third length ranges between 1.3 and 2.0. In some implementations, the first n-type FET and the second n-type FET share a p-type well, and the first p-type FET and the second p-type FET share an n-type well.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:
- a first device comprising:
  - a first plurality of fins extending in parallel along a first direction and comprising a first channel region and a first source/drain region,
  - a first gate structure wrapping over each of the first plurality of fins, and
  - a first source/drain contact disposed over the first source/drain region; and
- a second device comprising:
  - a first fin extending lengthwise along the first direction and aligned with one of the first plurality of fins along the first direction, the first fin comprising a second channel region and a second source/drain region,
  - a second gate structure wrapping over the first fin, and
  - a second source/drain contact disposed over the second source/drain region,
- wherein the first source/drain contact comprises a first length along a second direction perpendicular to the first direction,
- wherein the second source/drain contact comprises a second length along the second direction, and
- wherein a ratio of the first length to the second length is between about 1.1 and about 2.0.

2. The semiconductor structure of claim **1**,
- wherein the first device and the second device are disposed over a semiconductor substrate, and
- wherein the first plurality of fins and the first fin are disposed over a p-type well over the semiconductor substrate.

3. The semiconductor structure of claim **1**, wherein the first device and the second device are separated by an isolation feature extending lengthwise along the second direction.

4. The semiconductor structure of claim **3**, wherein the isolation feature comprises a dummy dielectric gate.

5. The semiconductor structure of claim **3**, wherein the isolation feature comprises a fin cutout region.

6. The semiconductor structure of claim **3**,
- wherein the isolation feature comprises an isolation p-type field effect transistor (FET) and an isolation n-type FET,
- wherein the isolation p-Type FET includes a first isolation gate structure electrically connected to a high voltage line, and
- wherein the isolation n-type FET includes a second isolation gate structure electrically connected to a low voltage line.

7. The semiconductor structure of claim **6**, wherein the first isolation gate structure is aligned with the second isolation gate structure along the first direction.

8. The semiconductor structure of claim **3**, further comprising:
- a third device comprising:
  - a second plurality of fins extending in parallel along the first direction and comprising a third channel region, and
  - the first gate structure wrapping over each of the second plurality of fins; and
- a fourth device comprising:
  - a second fin extending lengthwise along the first direction and aligned with one of the second plurality of fins along the first direction, and
  - the second gate structure wrapping over the second fin.

9. The semiconductor structure of claim **8**, wherein the third device and the fourth device are separated by the isolation feature.

10. The semiconductor structure of claim **8**,
- wherein the third device and the fourth device are disposed over a semiconductor substrate,
- wherein the second plurality of fins and the second fin are disposed over an n-type well over the semiconductor substrate.

11. A semiconductor structure, comprising:
- a semiconductor substrate having an n-type well and a p-type well;
- a first device disposed over the p-type well and comprising:
  - a first plurality of fins extending in parallel along a first direction, and
  - a first gate structure wrapping over each of the first plurality of fins;

a second device disposed over the p-type well and comprising:

a first fin extending lengthwise along the first direction and aligned with one of the first plurality of fins along the first direction, and

a second gate structure wrapping over the first fin;

a third device disposed over the n-type well and comprising:

a second plurality of fins extending in parallel along the first direction, and

the first gate structure wrapping over each of the second plurality of fins;

a fourth device disposed over the n-type well and comprising:

a second fin extending lengthwise along the first direction and aligned with one of the second plurality of fins along the first direction, and

the second gate structure wrapping over the second fin; and

an isolation feature extending between the first device and the second device as well as between the third device and the fourth device.

12. The semiconductor structure of claim **11**,

wherein the first plurality of fins comprise two fins, and wherein the second plurality of fins comprises two fins.

13. The semiconductor structure of claim **11**,

wherein the first plurality of fins comprise a first channel region and a first source/drain region adjacent the first channel region,

wherein the first gate structure is disposed over the first channel region,

wherein the first fin comprises a second channel region and a second source/drain region adjacent the second channel region, and

wherein the second gate structure is disposed over the second channel region.

14. The semiconductor structure of claim **13**,

wherein the first device further comprises a first source/drain contact disposed over the first source/drain region,

wherein the second device further comprises a second source/drain contact disposed over the second source/drain region,

wherein the first source/drain contact comprises a first length along a second direction perpendicular to the first direction,

wherein the second source/drain contact comprises a second length along the second direction, and

wherein a ratio of the first length to the second length is between about 1.1 and about 2.0.

15. The semiconductor structure of claim **11**, wherein the isolation feature comprises a dummy dielectric gate.

16. The semiconductor structure of claim **11**, wherein the isolation feature comprises a fin cutout region.

17. The semiconductor structure of claim **11**,

wherein the isolation feature comprises an isolation p-type field effect transistor (FET) and an isolation n-type FET,

wherein the isolation p-Type FET includes a first isolation gate structure electrically connected to a high voltage line, and

wherein the isolation n-type FET includes a second isolation gate structure electrically connected to a low voltage line.

18. The semiconductor structure of claim **17**, wherein the first isolation gate structure is aligned with the second isolation gate structure along the first direction.

19. A semiconductor structure, comprising:

a semiconductor substrate comprising a p-type well and an n-type well;

a first device disposed over the p-type well and comprising:

a first plurality of fins extending in parallel along a first direction and comprising a first channel region and a first source/drain region,

a first gate structure wrapping over each of the first plurality of fins, and

a first source/drain contact disposed over the first source/drain region; and

a second device disposed over the p-type well and comprising:

a first fin extending lengthwise along the first direction and aligned with one of the first plurality of fins along the first direction, the first fin comprising a second channel region and a second source/drain region,

a second gate structure wrapping over the first fin, and

a second source/drain contact disposed over the second source/drain region,

wherein the first source/drain contact comprises a first length along a second direction perpendicular to the first direction,

wherein the second source/drain contact comprises a second length along the second direction,

wherein a ratio of the first length to the second length is between about 1.1 and about 2.0.

20. The semiconductor structure of claim **19**, further comprising:

a third device disposed over the n-type well and comprising:

a second plurality of fins extending in parallel along the first direction and comprising a third channel region, and

the first gate structure wrapping over each of the second plurality of fins; and

a fourth device disposed over the n-type well and comprising:

a second fin extending lengthwise along the first direction and aligned with one of the second plurality of fins along the first direction, and

the second gate structure wrapping over the second fin.

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